

Hao Zhou

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SUMMARY

Motivated Master's student in Biostatistics with a robust foundation in **mathematics** and **statistics**, seeking to leverage expertise in **machine learning** and **data analytics** for advancing public health research. Experienced in developing predictive models and statistical analyses using **Python** and **R**.

EDUCATION

Tulane University	<i>New Orleans, LA</i>	Aug 2023 – Exp. May 2025
Master of Science in Biostatistics		GPA: 4.0/4.0
Coursework: Statistical Learning in Data Science, Categorical Data Analysis, Survival Data Analysis, Bayesian Inference, Regression Analysis, Principles of Sampling, Design of Experiments, Statistical Inference		
Nanjing University	<i>Nanjing, China</i>	Sept 2018 - June 2022
Bachelor of Science in Statistics		
Coursework: Computational Statistics, Statistical Machine Learning, Time Series Analysis, Multivariate Statistical Analysis, Stochastic Processes, Mathematical Statistics, Foundations of Probability Theory, Numerical Methods and Experiment, Introduction to Database, Mathematical Analysis, Higher Algebra		

RESEARCH

Tulane University, Department of Biostatistics and Data Science	<i>New Orleans, LA</i>
Research Assistant	Apr 2024 - Present

- Explored Multiple Instance Learning (MIL) applications in public health by comparing several data aggregation methods for multiple voice recordings to predict Parkinson's disease. Achieved the highest accuracy of 0.92 and MCC of 0.78 on the test dataset.
- Designed and implemented a comprehensive machine learning pipeline in Python for processing and analyzing public health datasets, which streamlined data preprocessing, feature engineering, model selection, hyperparameter tuning, and model evaluation.
- Developed a Quantitative Structure–Activity Relationship (QSAR) pipeline utilizing seven machine learning algorithms to model anti-aggregative activity of quinoline analogs, with the neural network model achieving the best performance and an R^2 of 0.90 on the test dataset.

CONFERENCE PRESENTATION

Drug Discovery & Development Colloquium/MALTO 2024	<i>Little Rock, AR</i>	June 2024
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- **Workshop:** Kakraba, S., Yang, Z., **Zhou, H.** (2024). A Python-based Machine Learning Pipeline for Predictive Modeling.
- **Poster:** **Zhou, H.**, Kakraba, S., Mainali, N., Bowroju, S. K., Johnson, J., Ayyadevara, S., ... & Reis, R.J.S. (2024). Modeling Anti-Aggregative Activity of Quinoline Analogs Using Machine Learning-Oriented Quantitative Structure-Activity Relationship. *Drug Discovery & Development Colloquium/MALTO 2024*.

PUBLICATIONS

* Indicates co-first authorship.

1. Yang, Z.*, **Zhou, H.***, Srivastav, S., Shaffer, J. G., Abraham, K. E., Naandam, S. M., Kakraba, S. Optimizing Parkinson's Disease Prediction: A Comparative Analysis of Data Aggregation Methods Using Multiple Voice Recordings via an Automated Artificial Intelligence Pipeline. Submitted for publication.
2. Tran, K., Nguyen, P., Opeoluwa, A., Okakuro, E., Huang, M., Liu, H., Yang, Z., **Zhou, H.**, Dang, N. (In

press). Prognostic Factors in Medulloblastoma: A SEER Database Analysis of Survival Outcomes from 2000 to 2020. *The American Journal of the Medical Sciences*.

3. Nguyen, P., Tran, K., Opeoluwa, A., Okakuro, E., Huang, M., Liu, H., Yang, Z., **Zhou, H.**, Dang, N. (In press). Survival Analysis of Rhabdomyosarcoma: A SEER Population-Based Study of 79 Cases. *The American Journal of the Medical Sciences*.
4. Huang, M., Tran, K., Nguyen, P., Opeoluwa, A., Okakuro, E., Liu, H., Yang, Z., **Zhou, H.**, Dang, N. (In press). Population-Based Survival Analysis of Ocular Lymphoma: Insights from SEER Database (2000–2020). *The American Journal of the Medical Sciences*.

SKILLS

Programming and Tools: Python, R, SQL, SAS, PyTorch

PROJECTS

Bird Image Classification using Convolutional Neural Network

- Built a Convolutional Neural Network (CNN) model using PyTorch to classify 90K images spanning 525 bird species.
- Implemented Data Augmentation and Data Normalization.
- Utilized pretrained models including AlexNet and ResNet, fine-tuning them to adapt to the output requirements for the classification task.
- Evaluated the model on test data and identified the best model with an accuracy of 98%.

Anime Recommendation System based on Matrix Factorization

- Implemented a recommendation model in PySpark to suggest anime likely to interest users.
- Conducted Online Analytical Processing (OLAP) with Spark SQL.
- Trained Alternating Least Squares (ALS) model for Matrix Factorization to provide personalized recommendations, and developed user-based and item-based algorithms to handle cold start problems.
- Tuned model hyperparameters through cross validation and achieved the optimal model with an RMSE of 1.2 in a user rating range of 1-10.

Natural Language Processing on ChatGPT Prompts

- Clustered ChatGPT prompts into groups, unveiling common scenarios where users engage with the model.
- Preprocessed the text data through tokenization, stemming, and TFIDF feature extraction.
- Employed K-means clustering and Latent Dirichlet Analysis (LDA) for unsupervised learning, identifying different prompt patterns and topics.
- Applied t-SNE for dimensionality reduction and visualized clustering results.

Stock Price Forecast Based on Time Series Analysis

- Developed a model in Python to predict stock prices for the upcoming week.
- Processed over four years of historical data and established dummy variables to account for the impact of the COVID-19 pandemic.
- Trained models including ARIMA and XGBoost to forecast stock price for a week.
- Evaluated model performance on test data with a minimal RMSE of 2.7.